

DF — Developing Fuels

OCR B (Salters) H433 storyline **DF**. The chemistry of car fuels: thermochemistry and enthalpy cycles, alkanes, alkenes and alcohols, cracking, electrophilic addition and addition polymers, gas volume calculations, isomerism and pollutants.

1. Enthalpy Changes

Standard Conditions and Notation

Standard conditions: A pressure of 100 kPa (approximately 1 atm), a stated temperature (usually 298 K / 25 °C), and all substances in their standard states (most stable form under these conditions). The standard enthalpy change is denoted ΔH° .

Enthalpy change (ΔH): The heat energy transferred in a reaction at constant pressure. Measured in kJ mol^{-1} .

Exothermic reaction: A reaction that releases heat energy to the surroundings. ΔH is negative. The products are at a lower energy level than the reactants.

Endothermic reaction: A reaction that absorbs heat energy from the surroundings. ΔH is positive. The products are at a higher energy level than the reactants.

Standard Enthalpy Definitions

Enthalpy Change	Symbol	Definition
-----------------	--------	------------

Formation	$\Delta_f H^\circ$	The enthalpy change when one mole of a compound is formed from its elements in their standard states under standard conditions
Combustion	$\Delta_c H^\circ$	The enthalpy change when one mole of a substance is completely burned in excess oxygen under standard conditions
Neutralisation	$\Delta_{\text{neut}} H^\circ$	The enthalpy change when an acid and a base react to form one mole of water under standard conditions
Atomisation	$\Delta_{\text{at}} H^\circ$	The enthalpy change when one mole of gaseous atoms is formed from the element in its standard state
Bond dissociation	D or $\Delta_{\text{bond}} H^\circ$	The enthalpy change when one mole of a specific covalent bond is broken homolytically in the gaseous state

Exam Tip: The standard enthalpy of formation of any element in its standard state is zero by definition. This is essential for Hess's law calculations using formation enthalpies. Always learn these definitions precisely — examiners will check for "one mole," "standard states," and "standard conditions."

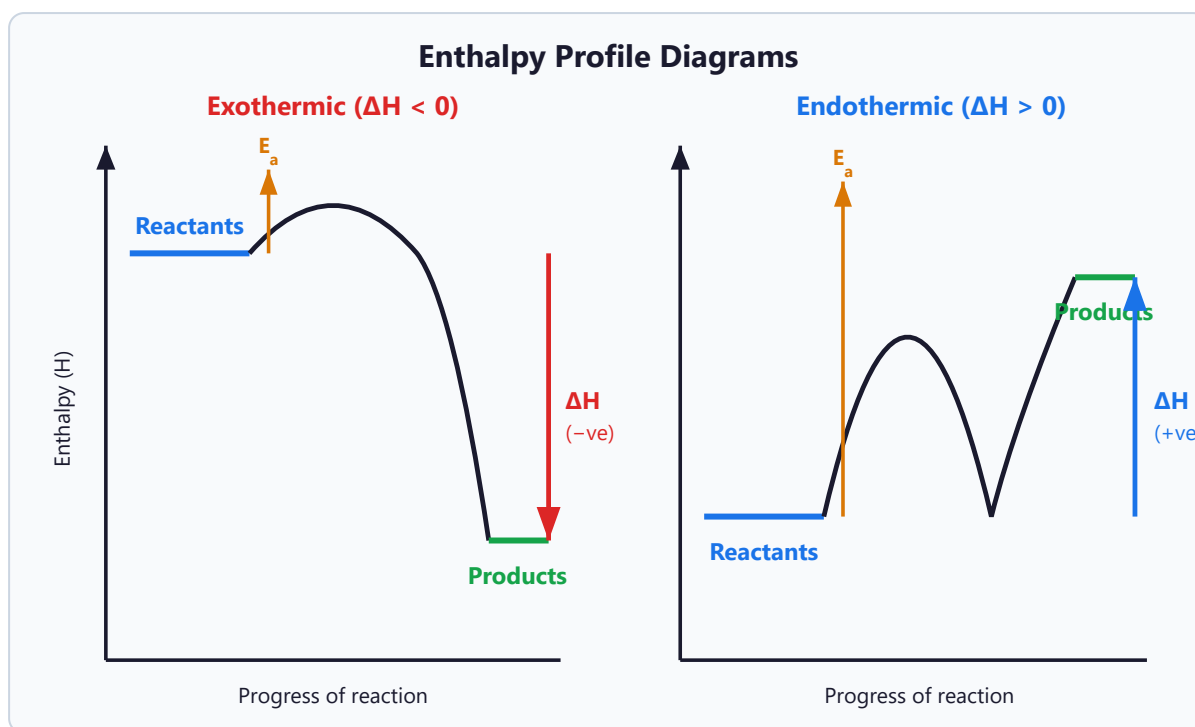
Worked Example — Calorimetry:

50.0 cm³ of 2.00 mol dm⁻³ HCl was added to 50.0 cm³ of 2.00 mol dm⁻³ NaOH. The temperature rose by 13.5 °C.

$$q = mc\Delta T = 100.0 \times 4.18 \times 13.5 = 5643 \text{ J} = 5.64 \text{ kJ}$$

$$n(\text{H}_2\text{O formed}) = 2.00 \times 0.0500 = 0.100 \text{ mol}$$

$$\Delta_{\text{neut}} H = -5.64 / 0.100 = \textbf{-56.4 kJ mol}^{-1} \text{ (negative because exothermic)}$$



Calorimetry

Calorimetry measures enthalpy changes experimentally by measuring temperature changes of a known mass of water (or solution).

$$q = mc\Delta T$$

where q is the heat energy transferred (in J), m is the mass of water/solution (in g), c is the specific heat capacity ($4.18 \text{ J g}^{-1} \text{ K}^{-1}$ for water), and ΔT is the temperature change (in K or $^{\circ}\text{C}$).

To find the molar enthalpy change: $\Delta H = -q / n$, where n is the number of moles of the limiting reagent. The negative sign is because heat gained by the surroundings (water) is heat lost by the system (reaction).

Sources of error in calorimetry include: heat loss to the surroundings (the most significant source), incomplete combustion, the assumption that the solution has the same specific heat capacity and density as pure water, and heat absorbed by the calorimeter itself.

Hess's Law

Hess's law: The total enthalpy change for a reaction is independent of the route taken, provided the initial and final conditions are the same. This is a consequence of conservation of energy.

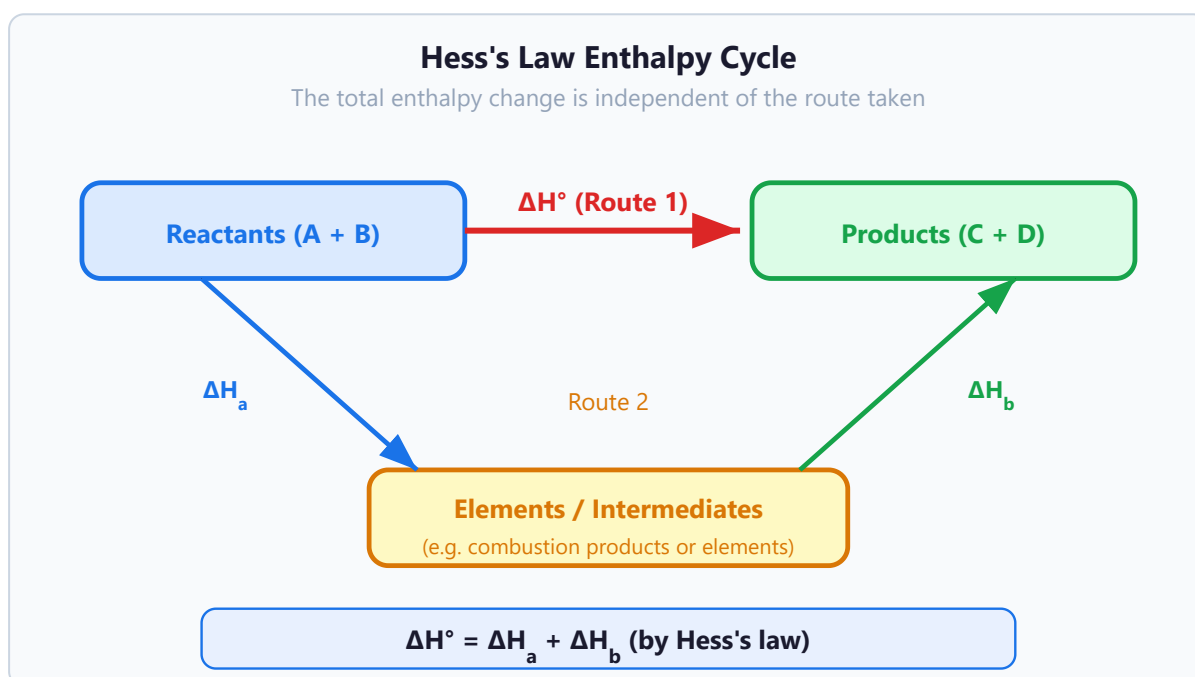
Hess's law can be applied using enthalpy cycles. Two common approaches:

Using enthalpies of formation:

$$\Delta_r H^\circ = \Sigma \Delta_f H^\circ(\text{products}) - \Sigma \Delta_f H^\circ(\text{reactants})$$

Using enthalpies of combustion:

$$\Delta_r H^\circ = \Sigma \Delta_c H^\circ(\text{reactants}) - \Sigma \Delta_c H^\circ(\text{products})$$



Bond Enthalpies

Mean bond enthalpy is the average energy required to break one mole of a particular type of bond across a range of different molecules. The enthalpy change of a reaction can be estimated:

$$\Delta H \approx \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$$

Breaking bonds is endothermic (energy input required); forming bonds is exothermic (energy released). This calculation uses mean bond enthalpies and gives only an estimate because actual bond energies vary depending on the molecular environment.

Exam Tip: Bond enthalpy calculations apply only to reactions where all reactants and products are in the gaseous state. If substances are liquids or solids, additional enthalpy changes (e.g. vaporisation) would need to be included.

2. Nomenclature and Functional Groups

IUPAC Nomenclature

Organic compounds are named systematically using IUPAC rules. The name consists of a root (indicating the longest carbon chain), a prefix (indicating substituents), and a suffix (indicating the principal functional group). Numbers (locants) indicate the positions of branches and functional groups on the chain, chosen to give the lowest possible set of numbers.

Carbon chain length	Root name
1	meth-
2	eth-
3	prop-
4	but-
5	pent-
6	hex-

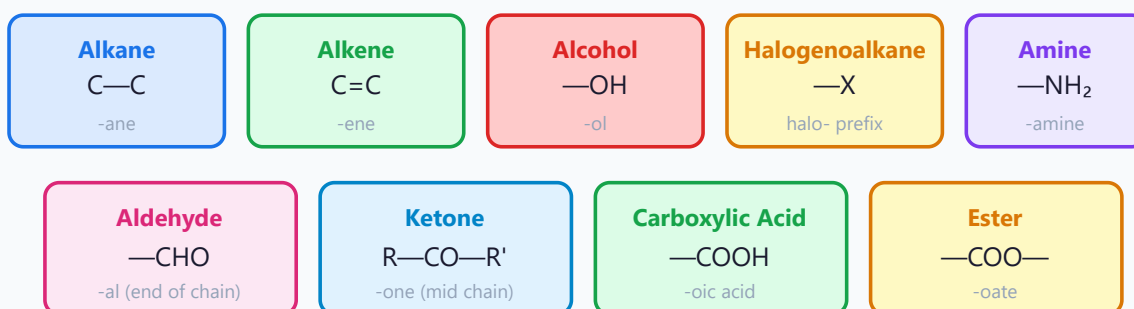
Key Functional Groups

Functional Group	General Formula	Suffix / Prefix	Example
Alkane	C_nH_{2n+2}	-ane	Ethane, C_2H_6
Alkene (C=C)	C_nH_{2n}	-ene	Propene, C_3H_6

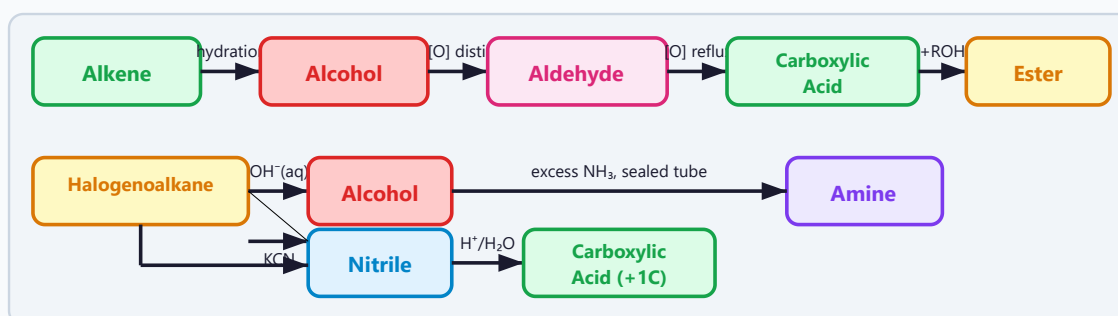
Alcohol (–OH)	$C_nH_{2n+1}OH$	-ol	Ethanol, C_2H_5OH
Halogenoalkane (–X)	$C_nH_{2n+1}X$	halo- prefix	Chloroethane, C_2H_5Cl
Aldehyde (–CHO)	RCHO	-al	Ethanal, CH_3CHO
Ketone (C=O)	RCOR'	-one	Propanone, CH_3COCH_3
Carboxylic acid (–COOH)	RCOOH	-oic acid	Ethanoic acid, CH_3COOH
Ester (–COO–)	RCOOR'	-oate	Ethyl ethanoate
Amine (–NH ₂)	RNH ₂	-amine	Ethylamine, $C_2H_5NH_2$

Homologous series: A family of organic compounds with the same functional group and general formula, in which successive members differ by CH_2 . Members have similar chemical properties and show a gradual trend in physical properties.

Key Functional Groups in Organic Chemistry



Key Interconversions



3. Isomerism

Structural Isomerism

Structural isomers: Molecules with the same molecular formula but different structural formulae (different arrangement of atoms). Types include chain isomerism (different carbon skeletons), position isomerism (same functional group at different positions), and functional group isomerism (different functional groups, e.g. an alcohol and an ether).

E/Z (Geometric) Isomerism

E/Z isomerism: A type of stereoisomerism that arises in molecules with a C=C double bond where each carbon of the double bond is attached to two different groups. The restricted rotation around the double bond means the groups can be arranged differently in space.

Z (zusammen): The higher-priority groups (assigned by Cahn-Ingold-Prelog rules based on atomic number) are on the same side of the double bond.

E (entgegen): The higher-priority groups are on opposite sides.

Optical Isomerism

Optical isomers (enantiomers): Non-superimposable mirror images of each other, arising when a carbon atom is bonded to four different groups (a chiral centre / asymmetric carbon). Enantiomers have identical physical and chemical properties except that they rotate the plane of plane-polarised light in opposite directions and may react differently with other chiral molecules (important in biological systems).

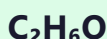
A **racemic mixture (racemate)** contains equal amounts of both enantiomers and shows no overall rotation of plane-polarised light because the rotations cancel out. Many synthetic reactions produce racemic mixtures, whereas enzymes typically produce a single enantiomer.

Exam Tip: To identify a chiral centre, look for a carbon atom bonded to four different groups. Draw the two mirror images to show the enantiomers. Remember that optical isomerism is a type of stereoisomerism, not structural isomerism.

Types of Formula — Ethanol (C₂H₆O)

Molecular Formula

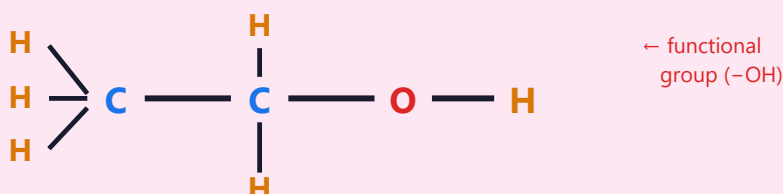
Actual number of each atom

Empirical Formula

Simplest whole-number ratio

Structural Formula

Shows groups but not bonds

Displayed Formula — shows every bond

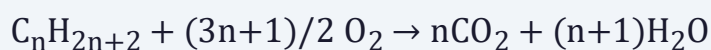
Skeletal formula: Each vertex/end = carbon; H atoms on C not shown; heteroatoms (O, N) shown explicitly

4. Alkanes

Alkanes are saturated hydrocarbons with only C–C single bonds and C–H bonds. They are relatively unreactive because the C–C and C–H bonds are strong and non-polar (or only very slightly polar).

Combustion

Alkanes undergo combustion in oxygen. Complete combustion produces carbon dioxide and water and is highly exothermic, making alkanes excellent fuels:

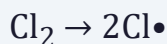


Incomplete combustion (in limited oxygen) produces carbon monoxide (toxic) and/or carbon (soot) instead of CO₂.

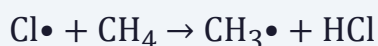
Radical Substitution

Alkanes react with halogens (Cl_2 , Br_2) in the presence of UV light via a free radical substitution mechanism. For methane and chlorine:

Initiation: UV light causes homolytic fission of the $\text{Cl}-\text{Cl}$ bond to form two chlorine radicals.



Propagation: Chain reactions where a radical reacts with a molecule to form a new radical.



Termination: Two radicals combine, removing radicals from the system.



Limitations: a mixture of products is formed (mono-, di-, tri-substituted etc.) because further substitution can occur. The reaction is not selective.

5. Alkenes

Alkenes are unsaturated hydrocarbons containing a $\text{C}=\text{C}$ double bond. The double bond consists of a sigma (σ) bond and a pi (π) bond. The π bond is a region of high electron density above and below the plane of the molecule, making alkenes susceptible to attack by electrophiles.

Electrophilic Addition

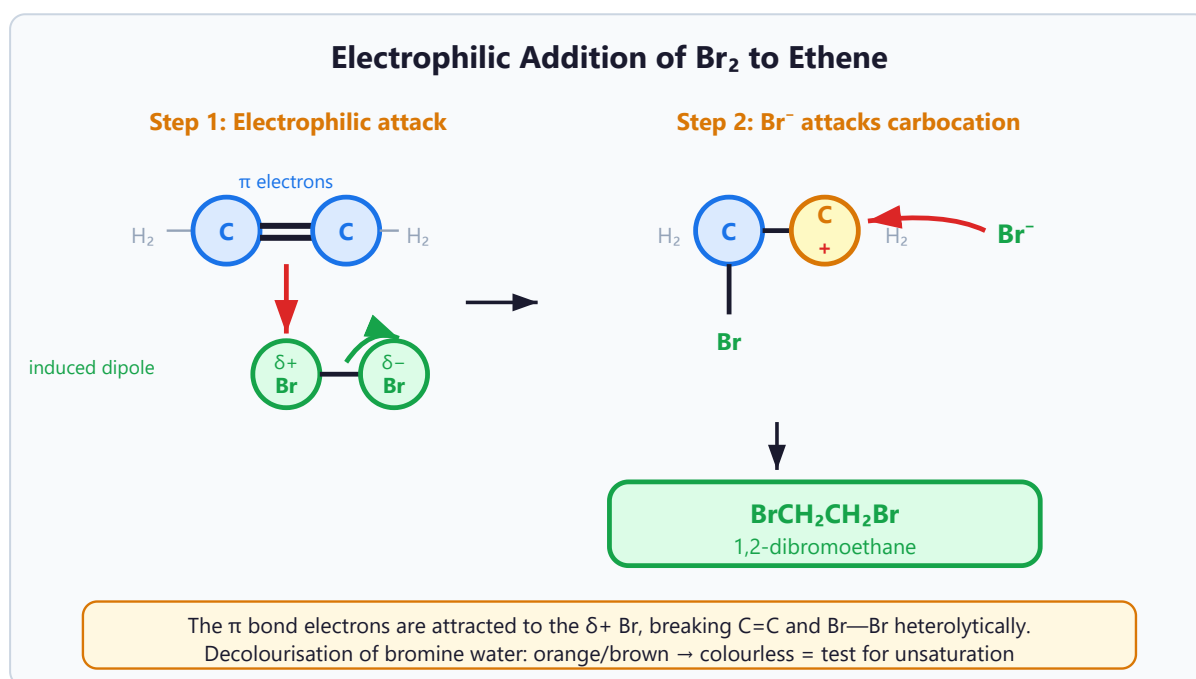
Electrophile: An electron-pair acceptor; a species attracted to regions of high electron density.

Examples: H^+ , NO_2^+ , Br_2 (when polarised).

Electrophilic addition: A reaction in which an electrophile is attracted to the electron-rich $\text{C}=\text{C}$ double bond, and atoms/groups add across the double bond, breaking the π bond.

Reaction with bromine (Br_2): The test for unsaturation. Bromine water is decolourised from orange/brown to colourless. The π electrons in the $\text{C}=\text{C}$ bond induce a dipole in the approaching Br_2 molecule. The δ^+ bromine atom acts as the electrophile, forming a bond with one carbon while the $\text{Br}-\text{Br}$ bond breaks heterolytically. A carbocation intermediate forms, which is then attacked by the Br^- ion.

Reaction with hydrogen halides (HBr): The H^+ (the electrophile) adds to one carbon of the double bond, and the halide ion adds to the other. Markovnikov's rule states that the hydrogen adds to the carbon already bonded to more hydrogen atoms, forming the more stable (more substituted) carbocation intermediate.



Reaction with concentrated sulfuric acid: H_2SO_4 adds across the double bond. Subsequent hydrolysis with water produces an alcohol (indirect hydration).

Reaction with steam (hydration): In the presence of an acid catalyst (H_3PO_4) at high temperature and pressure, steam adds across the double bond to form an alcohol directly. This is the industrial method for producing ethanol from ethene.

Polymerisation

Addition polymerisation: Many alkene monomers join together by breaking their $\text{C}=\text{C}$ double bonds and forming new $\text{C}-\text{C}$ single bonds. No other molecules are produced (100% atom economy). The polymer is named poly(monomer), e.g. poly(ethene), poly(propene), poly(chloroethene) (PVC).

Addition polymers are generally non-biodegradable because the C–C backbone is very stable and resistant to attack by microorganisms or enzymes.